The Brazilian Geodetic System — Part 1: SIRGAS2000

How Brazil changed its reference system: the inheritance, the design, the campaign and the adoption — four classes

---

Coordinate Systems Simulator

What we will see

1. Where we left off
2. What the SGB is
3. Measuring a country mark by mark
4. The topocentric datum
5. The cost of the inheritance
6. Wrap-up
7. Class 2: The Design (1993–1995)
8. Class 3: The Realization (May 2000)
9. Class 4: Adoption, Densification and Lifetime

This is Part 1

Position: how Brazil knows *where* each point is, and how that changed. **Height** (the leveling network) is left for Part 2.

Four classes

Class 1 · The inherited problem

How a whole country was measured before satellites — and why the result aged badly.

Class 2 · The design

1993–1995: how a reference system is planned for a continent, and the dress rehearsal called SIRGAS95.

Class 3 · The realization

May 2000: 184 stations measured at the same time, and the computation that turns them into coordinates.

Class 4 · Adoption and lifetime

The law, the conversion formula, the continuous network — and why the official coordinate no longer matches the receiver.

Each class is 50 minutes long, starts by recalling the previous one and ends with a summary. We will go **slowly**: one idea per slide.

Class 1

The inherited problem

class 1 of 4

Where we left off

Today's questions

1. What is the Brazilian Geodetic System, and what is it made of?
2. How was a whole country measured before GPS?
3. What is a "topocentric" datum, and why does it age badly?
4. What error did the inherited network carry — and how was it found?

Deal

By the end of the class you will be able to answer all four — in your own words.

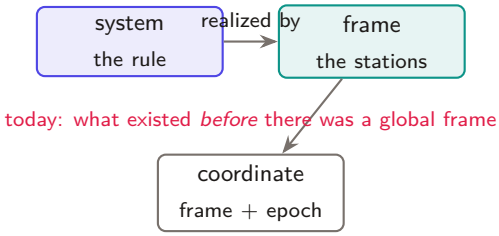
They come back on the last slide.

What the previous class left us

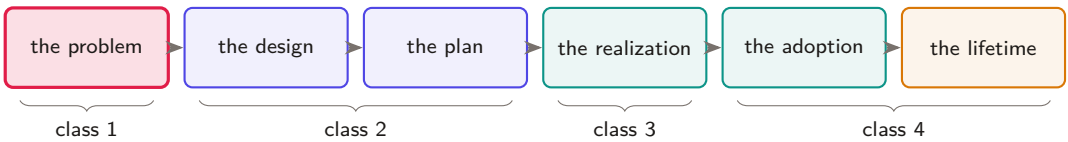
In the terrestrial reference systems class, three ideas:

1. the **system** is the written rule; you cannot measure anything with it;
2. the **frame** is a set of real stations that obey the rule with some small error;
3. a coordinate must say **in which frame and at which epoch**.

There we saw the lamp post that "moves" tens of meters between the old plan and today's map. Here we open that box: where did that shift come from?



The thread of the class



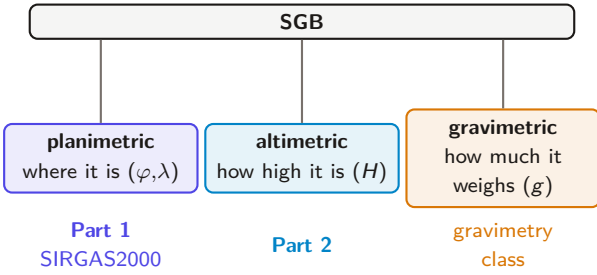
Today is only the first box — but it gives meaning to all the others: if the problem is not clear, SIRGAS2000 looks like a whim.

What the SGB is

The Brazilian Geodetic System, in one sentence

SGB

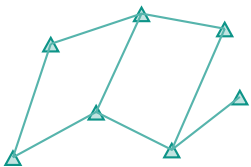
The set of **stations and networks** that gives the country its official **coordinates**, **heights** and **gravity** — established and maintained by IBGE.



It is not just a list of numbers: it is the *base* on which the map, the land register, the construction and your GPS stand.

By law: Decree 5.334/2005 says that the planimetric and altimetric references of the cartography are those of the SGB, *as established by IBGE*.

Two families of stations: passive and active



passive: marks in the ground
that someone must go and occupy



active: receivers running
24 hours a day, every day

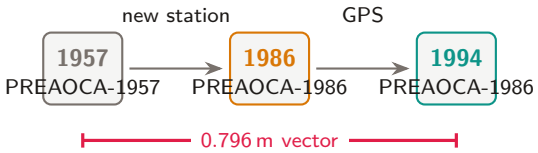
The **passive** network is the inheritance of this chapter: triangulation vertices (VT) and traverse stations (EP). The **active** ones are the RBMC (continuous GNSS receivers) and the RMPG (tide gauges), which come back in Class 4 and in Part 2.

One mark, three plates

The vertex **PREAOCA** was set in **1957**: a metal plate fixed straight into the rock, engraved `PREAOCA-1957`.

In **1986** a new occupation found the plate engraved `PREAOCA-1986`: another station had been set, with the same name.

In **1994** GPS reoccupied the 1986 plate — and the coordinate obtained differed from the old one by 0.796 **m**.



IBGE dropped this connection from the adjustment

Lesson

The **frame** is made of **real things**: they get replaced, destroyed and mixed up. (Two Brazilian SIRGAS stations were destroyed before being tied back to the network.)

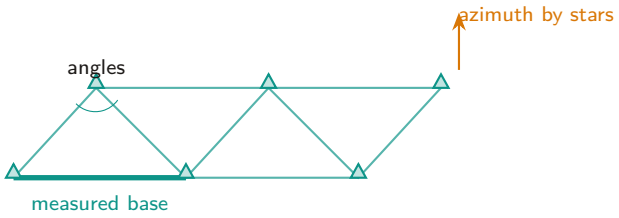
Measuring a country mark by mark

Triangulation: one ruler and many angles

Measuring a long distance on the ground is expensive and slow. Measuring *angles* is much easier.

So: measure **one base** carefully and from then on only angles. With the law of sines each triangle gives the next side — and the chain moves on.

To keep the chain from rotating, its orientation is anchored with **astronomical azimuths** (by the stars).



What went into the computation

observation	weight (σ)	in the 2006 adjust
horizontal direction	$3\,\mathrm{ppm}$	1
distance	$1\,\mathrm{ppm}$	
astronomical azimuth	$1\,\mathrm{ppm}$	389

Stations of the classical network

4774

triangulation vertices and traverse stations, in 2006

Almost everything was **directions**: angles between targets. Distances were few and orientation came from a short list of azimuths.

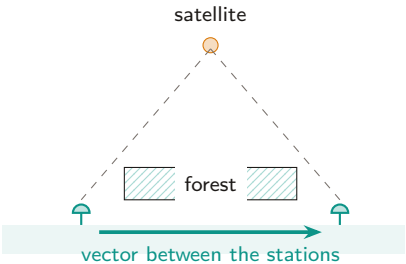
"$3\,\mathrm{ppm}$" reads *3 parts per million*: 3 mm of error per 1 km.

Where triangulation does not reach

In the Amazon the forest blocks the lines of sight: you cannot see the next mark.

Before GPS the way out was the **Doppler satellites**. Two receivers on the ground listen to the *same* satellite at the same time; what they share cancels, and the **vector between them** is left. This is **translocation**.

The 1996 network already had 179 Doppler stations and 15 from the Amazon translocation network.

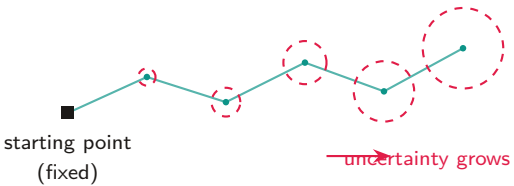


The error piles up

Each triangle inherits the position of the previous one. A small angle error at the start becomes a larger error *later*, because the next triangle starts from an already shifted point.

Result: the error **grows with the distance** from the starting point.

In a country 4 000 km across, the starting point is *very* far from the borders.



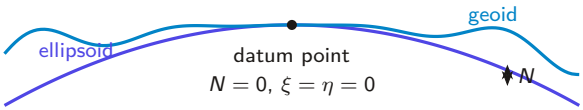
The topocentric datum

Pinning the ellipsoid to the Earth

To compute on the network we need a smooth, simple surface: the **ellipsoid**. The real Earth, with its lumps, resembles the **geoid**.

The two do not coincide. So one has to choose where to touch one against the other.

At the **datum point** we impose: the ellipsoid touches the geoid ($N = 0$, and the vertical coincides with the normal: $\xi = \eta = 0$). Far from it, the two drift apart.



schematic, not to scale

Córrego Alegre: chosen by the deflections of the vertical

While setting the triangulation chain in Santa Catarina, deflections of the vertical were measured **astronomically**. They leaned east: more mass to the west, less to the east. Conclusion: the datum point would sit better on the **plateau**.

The **Córrego Alegre** vertex and Hayford's international ellipsoid were chosen. Position and orientation at the point were determined **by the stars**. The computations were done with mechanical calculators and logarithm tables.

Córrego Alegre datum

ellipsoid: **Hayford 1924**

$a = 6378388\,\mathrm{m} \quad 1/f = 297$

datum point: Córrego Alegre vertex

$\varphi = -19^\circ 50' 14{,}91''$

$\xi = \eta = 0, \quad N = 0\,\mathrm{m}$

From Córrego Alegre to SAD69

1. To have *one* reference for South America, with support from the American mapping agency, vertices in several countries are measured astronomically: the **PSAD56** is born, with origin at La Canoa, Venezuela. A trilateration network (HIRAN) links the Brazilian and Venezuelan chains.
2. The fit to the geoid there was poor. 2113 gravimetric stations were measured around a new candidate: the **Chuá** vertex.
3. With it, the **SAD69** (*South American Datum 1969*).

SAD69 datum

ellipsoid: **UGGI67**

$a = 6378160\,\text{m} \quad 1/f = 298.25$

datum point: Chuá vertex

$\xi = 0.31'', \quad \eta = -3.52'', \quad N = 0\,\text{m}$

Notice

To choose where to pin the ellipsoid, **gravimetry** was used — the gravimetry class shows up again.

Three references side by side

	Córrego Alegre	**SAD69**	**SIRGAS2000**
ellipsoid	Hayford 1924	UGGI67	GRS80
a (m)	6378388	6378160	6378137
$1/f$	297	298.25	298.257222101
origin	datum point on the ground	datum point on the ground	Earth's center of mass
how it is fixed	stars	stars + gravimetry	satellites

The first two columns are topocentric: the "center" is a point on the ground in Minas Gerais.
The third is **geocentric**.

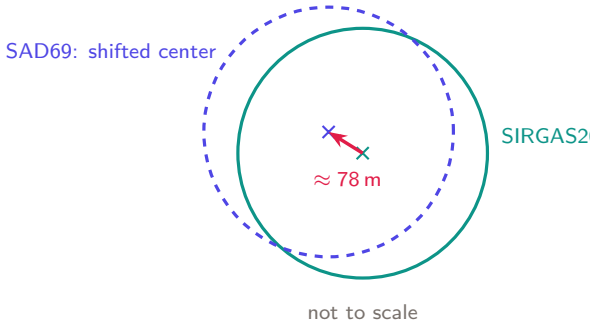
Topocentric = center out of place

Because the SAD69 was pinned at a point on the ground, the center of its ellipsoid does not coincide with the Earth's center of mass.

By the official parameters (3 translations), the distance between the two centers is:

$$\sqrt{67{,}35^2 + 3{,}88^2 + 38{,}22^2} \approx \mathbf{77.5\,m}$$

About eighty meters. And everything measured *from* that center inherits the offset.



Stop and think

A question

The two centers are 77.5 m apart. But the post only "moves" between 57 and 76 m on the Brazilian ground.

Why isn't the horizontal shift always 77.5 m?

place	ΔN	ΔE	Δh
Fortaleza	-41.7	-38.9	-29.5
Curitiba	-54.2	-48.5	-3.0
Chuí	-55.2	-51.8	$+8.1$

meters, SAD69 $\rightarrow$ SIRGAS2000, computed with the 3 official parameters. *Always* to the southwest.

Stop and think

A question

The two centers are 77.5 m apart. But the post only "moves" between 57 and 76 m on the Brazilian ground.

Why isn't the horizontal shift always 77.5 m?

Answer

The 77.5 m vector has a fixed direction in space. At each place, a part of it points along the vertical (it becomes *height*) and only the rest is *horizontal*. Because the vertical changes from place

place	ΔN	ΔE	Δh
Fortaleza	-41.7	-38.9	-29.5
Curitiba	-54.2	-48.5	-3.0
Chuí	-55.2	-51.8	$+8.1$

meters, SAD69 $\rightarrow$ SIRGAS2000, computed with the 3 official parameters. *Always* to the southwest.

In Fortaleza a good part of the vector becomes height; in the south, almost none.

The cost of the inheritance

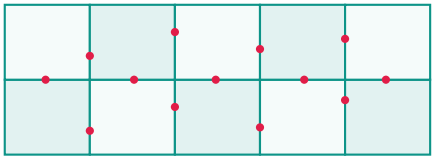
The network, stitched in pieces

The first computer adjustment of the SAD69 (American mapping agency, program HAVOC) split the Brazilian network into **10 areas**, computed separately: the computer of the time could not take more.

In the 1980s IBGE went on like this: each new area was computed *holding* the junction stations at the coordinates of the previous computation.

Each seam keeps its neighbor's error. IBGE itself records that this introduced distortions.

10 areas computed one at a time



- junction stations: the error passes from one area to the

1996: the first global adjustment

In 1996 IBGE adjusted *everything at once*, for the first time. Before, there were only partial adjustments.

classical network stations	4760
GPS stations	187
Doppler stations	179

Result published in September 1996 in the Geodetic Database (BDG), the official catalog of the SGB.

What IBGE concluded

For the first time a "consistent picture of the quality of the network" was obtained.

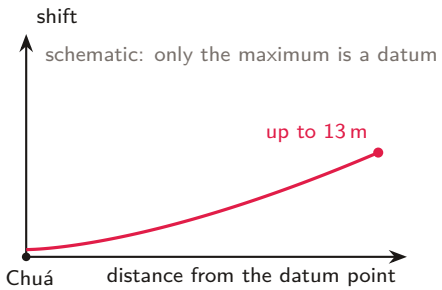
And what it saw

Compared with what was in the BDG, the horizontal shift **increases with the distance from the datum point**, reaching 13 **m** at the northern, southern and western extremes.

The distortion grows with distance

It is the same picture as “the error piles up”, now at country scale.

Near Chuá the network is good: that is where the ellipsoid was pinned. Far from it, geoid and ellipsoid separate, and the piecewise adjustment piled up the rest.



Your turn

How many parts per million?

(a) One extreme of the country is about 2000 km from Chuá and piled up 13 m. How many mm per km is that (in ppm)?

(b) The classical network "promised" $3\,\mathrm{ppm}$ on directions. Over 2000 km, how much would that be?

Your turn

How many parts per million?

(a) One extreme of the country is about 2000 km from Chuá and piled up 13 m. How many mm per km is that (in ppm)?

(b) The classical network "promised" $3\,\mathrm{ppm}$ on directions. Over 2000 km, how much would that be?

Answers

(a) $\dfrac{13\,\mathrm{m}}{2\,000\,000\,\mathrm{m}} = \mathbf{6.5}\,\mathrm{ppm}$.

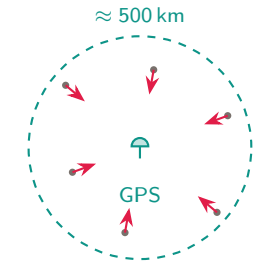
(b) $3 \times 10^{-6} \times 2000\,\mathrm{km} = \mathbf{6\,m}$.

The observed error is of the same order as the promised one: the network was not "bad", it was at the limit of what the technique allowed over such a distance.

GPS shows what the network was hiding

The classical surveys stopped in the late 1980s. In early 1991 IBGE bought **four GPS receivers**. By 2006 there were 1600 GPS stations.

In 2004 and 2005, 12 GPS campaigns were made *on top of* marks of the classical network. It was found that **a single GPS connection changes the coordinates of its neighbors by up to 9 m**, within a radius of about 500 **km**.



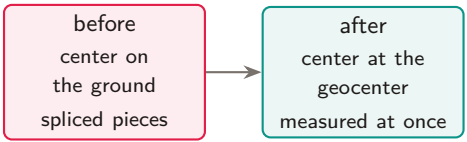
And now?

In 2000 IBGE creates the **PMRG** (Geodetic Reference Change Project), by resolution of the 1999 Brazilian Cartography Congress. The change is unavoidable. The question is: **to which reference?**

Wrap-up

"What if the center were the Earth's center, measured all at once by satellites?"

If the distortion comes from pinning the ellipsoid to a point on the ground and splicing pieces, the cure is to **pin it at the center of mass** and **measure everything at the same time**.



Class 2: who decided this, and how it is planned

Take-home — class 1

1. The **SGB** is the set of stations and networks that gives the country official position, height and gravity. The **passive** network (marks in the ground) is the inheritance; the **active** ones (RBMC, RMPG) are the present.
2. The country was measured by **triangulation**: one base, many angles, azimuths by the stars — and Doppler satellites where the forest got in the way. The error grows with distance from the starting point.
3. A **topocentric datum** pinned the ellipsoid to a point on the ground (Córrego Alegre, later Chuá). So the center of the SAD69 is about 78 **m** from the Earth's center.
4. The network was **spliced from pieces** because of computer limits. The 1996 global adjustment showed shifts of up to 13 m far from the datum point.
5. GPS connections on classical marks move their neighbors by up to 9 m, within a 500 km radius: **GPS does not just measure, it audits**.

Next class: in 1993, in Asunción, a group decides to pin the reference system at the Earth's center — and begins planning how.

Class 2: The Design (1993–1995)

Class 2 Hook: The Continent Without a Common Center

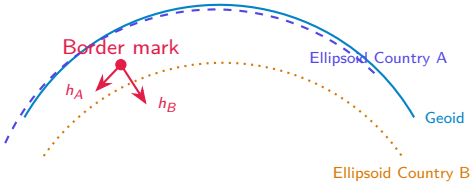
The question for this class: how do you plan a single geodetic reference system for an entire continent, when each country used a different ellipsoid and none of them was centered at the Earth's center?

- In Class 1 we saw that Brazil used **Córrego Alegre** and later **SAD69**, pinned to Minas Gerais with accumulated errors reaching 13 **m**.
- Meanwhile Argentina used *Campo Inchauspe*; Venezuela and Colombia used *La Canoa* (PSAD56); Chile used *SAD56*.
- At the border between two nations, the exact same bridge or dam had two official sets of coordinates differing by dozens or hundreds of meters!

The South American Patchwork in 1993

Border issues

- Border monuments surveyed by neighbouring countries disagreed.
- Each nation had its own reference ellipsoid (Hayford 1924, UGGI67, Clarke 1866).
- Satellite navigation systems (GPS) provided geocentric coordinates, conflicting with all national map sheets.



The same physical feature projected onto different surfaces with separate origins.

Asunción, October 1993: The Historic Turning Point

- In October 1993, the *International Conference for the Definition of a South American Geocentric Datum* took place in **Asunción (Paraguay)**.
- Jointly organized by:
  - **IAG** (International Association of Geodesy);
  - **IPGH** (Pan-American Institute of Geography and History);
  - **DMA** (*Defense Mapping Agency*, now NGA, USA).
- There **SIRGAS** was born: originally *South American Geocentric Reference System*.

The goal set in Asunción:

Adopt a **single geocentric** reference frame, compatible with ITRS and space techniques (GPS), serving as the foundation for all nations on the continent.

Name evolution: in 2001, the UN recommended extending the system to the Americas. In Nov/2020, it became *Geodetic Reference System for the Americas*.

The Initial Working Structure (1993)

Working Group I (WG I)

Reference System

- Define the continental geocentric reference system.
- Plan, execute and process the high-precision continental GPS network.
- Materialize the initial continental "backbone".

Working Group II (WG II)

Geocentric Datum

- Guide the integration of national geodetic networks into the new system.
- Densify stations within the borders of each member country.
- Define transformation parameters between classical datums and SIRGAS.

What about height? In 1993, the WGs handled only 3D geometry (X, Y, Z). The unification of physical heights was created in 1997 with **WG III** (the core focus of our Part 2!).

The Four Great Design Decisions

Instead of improvising, geodesists in Asunción made 4 fundamental decisions:

1. The System

ITRS / IERS

Global open scientific standard

2. The Ellipsoid

GRS80

IUGG 1979 geocentric parameters

3. The Technique

GPS espacial

Continuous simultaneous tracking

4. The Structure

Em camadas

Global → Continental → National

Let us inspect the logic behind each of these four choices.

Decision 1: Why ITRS from IERS?

- **What was chosen:** o *International Terrestrial Reference System* (ITRS), mantido pelo IERS.
- **Alternatives:**
  - Criar um sistema sul-americano próprio;
  - Adotar diretamente o WGS 84 militar.
- **Why ITRS?**
  - É o padrão científico mundial mais rigoroso da astronomia e geodésia.
  - Não depende de um único país nem de uma única agência militar.
  - Combina quatro técnicas espaciais independentes: VLBI, SLR, GNSS e DORIS.

ITRS definition:

– Origin at Earth's center of mass (including oceans and atmosphere);

– Metric scale in local relativistic terrestrial frame;

– Orientation aligned with BIH 1984.0 pole and meridian;

– Time evolution with no-net-rotation of the crust (*NNR* condition).

Stop and Think: Why Not Use WGS 84 Directly?

Common question: "GPS receivers output WGS 84 coordinates. Why did South America need SIRGAS instead of just decreeing WGS 84?"

Stop and Think: Why Not Use WGS 84 Directly?

Common question: "GPS receivers output WGS 84 coordinates. Why did South America need SIRGAS instead of just decreeing WGS 84?"

WGS 84 (Military)

- Controlled by the US Department of Defense.
- Lacks permanent physical monuments accessible to civilians in South America.
- Updates occur without prior consultation with other nations' civil agencies.

SIRGAS (Scientific / Sovereign)

- The **regional civil and open** realization of ITRS in the Americas.
- Materialized on marks and continuous stations run by national agencies (IBGE, etc.).
- Public, auditable data fully consistent with IGS.

Decision 2: The GRS80 Reference Ellipsoid

- ITRS is a 3D orthogonal Cartesian coordinate system (X, Y, Z).
- For mapping and engineering (φ, λ, h), a **biaxial ellipsoid of revolution** is associated.
- The ellipsoid adopted by SIRGAS is **GRS80** (*Geodetic Reference System 1980*):
  - Semi-eixo maior: $a = \mathbf{6378137}\ \mathrm{m}$;
  - Achatamento: $1/f = \mathbf{298.257222101}$;
  - Constante gravitacional geocêntrica: $GM = 3986005 \times 10^8\ \mathrm{m}^3/\mathrm{s}^2$;
  - Velocidade angular: $\omega = 7292115 \times 10^{-11}\ \mathrm{rad/s}$.

GRS80 versus WGS 84:

Both share the exact same semi-major axis ($a = 6378137\,\mathrm{m}$).

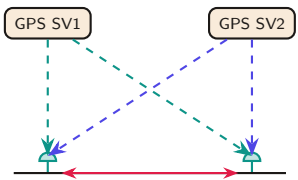
They differ only in flattening by 0.00000016 due to the J_2 harmonic definition.

On Earth's surface, this shift produces at most **0.1 mm** in ellipsoidal height — *negligible for practical surveying!*

Decision 3: The Satellite Surveying Technique

Why **GPS** instead of VLBI, SLR, or Doppler?

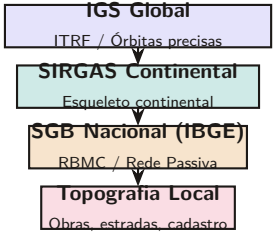
- **VLBI e SLR** were the most accurate globally, but required multi-ton radiotelescopes and giant lasers; unfeasible at 50 remote marks in the rainforest.
- **Doppler (TRANSIT)** was outdated by the 1990s, offering meter-level precision.
- **GPS** offered the ideal combination:
  - Antenas e receptores leves, transportáveis por equipe de campo;
  - Rastreio simultâneo da fase da onda portadora (L1 e L2);
  - Precisão relativa milimétrica em linhas de base de milhares de quilômetros.



Carrier phase + simultaneous tracking = cancellation of common errors.

Decision 4: Layered Hierarchy (The Subway Analogy)

- How do you link a street corner monument in rural Brazil to the Earth's center?
- You don't measure the corner mark from the North Pole! You use a **layered hierarchy**:
  1. **Layer 1 (Global):** Rede IGS mundial (dezenas de estações com VLBI/SLR e órbitas ultra-precisas).
  2. **Layer 2 (Continental):** Rede SIRGAS (esqueleto de estações de alta estabilidade amarradas ao IGS).
  3. **Layer 3 (National):** Redes do SGB / IBGE (RBMC contínua e marcos passivos espalhados pelo território).
  4. **Layer 4 (Local):** O topógrafo no campo amarrado ao marco ou à estação RBMC mais próxima.



Each layer transfers the accuracy of the higher tier to a denser scale.

SIRGAS95: The Continent's First Test

The May/June 1995 Campaign

- **Period:** 26 de maio a 4 de junho de 1995 (10 dias consecutivos de 24 horas).
- **Countries:** 11 países da América do Sul mobilizados.
- **Stations:** 58 estações principais + 9 excêntricas.
- **Brazil:** 11 estações fundamentais ocupadas pelo IBGE e parceiros.

- **Station selection criteria:**
  - Todos os observatórios espaciais existentes na América do Sul (SLR, VLBI, DORIS).
  - Distribuição geográfica homogênea pelo continente.
  - Estabilidade geológica comprovada (pilares de concreto sobre rocha sã).
  - Coincidência com marcos das redes clássicas existentes para calcular parâmetros de conversão.

The 11 Brazilian Stations in SIRGAS95

Strategically distributed to cover the entire national territory:

- **Fortaleza (CE):** sítio espacial global (VLBI / SLR / GPS).
- **Brasília (DF):** coração do planalto central.
- **Curitiba (PR):** Universidade Federal do Paraná (UFPR).
- **Presidente Prudente (SP):** UNESP.
- **Viçosa (MG):** Universidade Federal de Viçosa (UFV).
- **Belém (PA) e Manaus (AM):** nós amazônicos.
- **Bom Jesus da Lapa (BA):** interior nordestino.
- **Cuiabá (MT):** centro-oeste.
- **Santa Maria (RS):** extremo sul.
- **Cruzeiro do Sul (AC):** extremo oeste.

A historic leap:

For the first time, points over 3500 km apart in the Amazon and the southern plains were linked with uncertainty **under 1 cm** in the same frame!

Pioneering Challenge: Antenna Calibration

- In 1995, each nation used different brands of receivers and antennas: Ashtech, Leica, Trimble, and TurboRogue.
- The satellite signal is captured not at the antenna's physical tip, but at its **electronic phase center**.
- The phase center varies with satellite elevation and azimuth (*PCV*). Mixing uncalibrated antennas creates errors up to **5 to 10 cm** vertically!

The solution adopted in 1995:

9 eccentric stations were set up meters away from main marks using differing antenna models.

Direct short-baseline surveying calibrated relative phase center offsets across all brands used in the campaign.

SIRGAS95 Processing: Three Independent Centers

To ensure full scientific rigor and avoid bias, raw data were processed independently by three centers:

1. **DGFI** (*Deutsches Geodätisches Forschungsinstitut*, Germany) — Bernese scientific software;
2. **NIMA / DMA** (*National Imagery and Mapping Agency*, EUA) — GIPSY/OASIS scientific software;
3. **IBGE** (Brazil) — Bernese scientific software.

Free-network solutions were combined by least squares and tied to **ITRF94** at epoch **1995.4** via **7 global IGS stations** using a Helmert transformation.

The 7 1995 IGS Anchors

- **Fortaleza** (Brazil)
- **Kourou** (French Guiana)
- **Arequipa** (Peru)
- **Santiago** (Chile)
- **Easter Island** (Easter Island, Chile)
- **Galápagos** (Ecuador)
- **Pamatai** (French Polynesia)

Crustal Surprise: The Dynamic Earth!

- The SIRGAS95 campaign concluded on June 4, 1995 with sub-centimeter accuracy across the continent.
- Less than two months later, on **July 30, 1995**, the massive **Antofagasta earthquake** ($M_w = 8.0$) struck northern Chile!
- Geodetic monuments suddenly shifted by up to **70 cm**! The coordinates calculated in May no longer matched the rock in August!

The great lesson learned:

A modern geodetic frame cannot just be a list of static coordinates fixed forever.

The South American plate drifts at **1 to 2 cm/yr** northwestward, and the Andean margin deforms continuously.

Without velocity models, a high-precision frame ages quickly!

What Was Missing After 1995?

What worked in 1995

- Proved technical cooperation among 11 countries, military and civil agencies.
- Demonstrated sub-centimeter consistency with ITRF94.
- Proved GPS was the definitive tool for continental geodesy.

What was still missing

- **Density:** 58 stations were too few for an 18M km^2 continent.
- **Central America and Caribbean:** were outside the original scope.
- **Physical heights:** no formal tie between tide gauges and SIRGAS.

Thus arose the decision to conduct a new, massive campaign in the year **2000**, reoccupying old sites and connecting all tide gauges and networks across Latin America.

Your Turn: Calculating Tectonic Drift

Problem

A geodetic station in Curitiba has an estimated plate motion of $1.2\,\mathbf{cm/yr}$ northwestward.

(a) If we measure this mark today, how many centimeters has it shifted relative to its epoch 1995.4 coordinate after 10 years?

(b) If we ignore this shift in a high-precision survey in 2005, is the error noticeable?

Your Turn: Calculating Tectonic Drift

Problem

A geodetic station in Curitiba has an estimated plate motion of $1.2\,\mathbf{cm/yr}$ northwestward.

(a) If we measure this mark today, how many centimeters has it shifted relative to its epoch 1995.4 coordinate after 10 years?

(b) If we ignore this shift in a high-precision survey in 2005, is the error noticeable?

Step-by-step Solution

(a) $\Delta s = v \times \Delta t$

$= 1.2\,\mathrm{cm/ano} \times 10\,\mathrm{anos} = \mathbf{12\,cm}$.

(b) **Yes, it is huge!** For coarse decimeter mapping it might pass, but for geodetic control and engineering (1 to 2 cm), $12\,\mathbf{cm}$ ruins the adjustment!

Takeaways (Class 2)

1. **Asunción 1993** was the turning point: isolated national topocentric datums were abandoned in favor of a continental geocentric reference frame.
2. SIRGAS adopted **ITRS** as system, the **GRS80** ellipsoid, and **GPS** technology in a layered hierarchy.
3. **SIRGAS95** materialized the first backbone with 58 stations (11 in Brazil) and sub-centimeter accuracy, tied to ITRF94 at epoch **1995.4**.
4. The major lesson was that the **Earth is alive**: the crust deforms via plate tectonics and earthquakes, requiring velocity models and periodic campaigns.

Next class: the massive May 2000 campaign with 184 stations, and the adjustment mathematics that generated SIRGAS2000 at epoch 2000.4.

Class 3: The Realization (May 2000)

Class 3 Hook: From the Rainforest to Computation

The question for this class: how do 184 GPS receivers scattered across rainforests, deserts, islands, and capitals in two continents get turned into a **single set of millimeter-accurate coordinates**?

- In Class 2 we examined the design and the pilot SIRGAS95 campaign with 58 stations.
- In May 2000, the continent mounted an unprecedented scientific mobilization: 184 **stations tracking simultaneously day and night**.
- We will discover the logistics, the double-difference physics, and the mathematics of **least squares adjustment with constraints**.

The May 2000 Campaign: The Leap in Scale

The Continental Operation

- **Dates:** 10 a 19 de maio de 2000 (10 dias consecutivos).
- **Sessions:** 24 horas contínuas de rastreio diário com taxa de 30 segundos.
- **Coverage:** América do Sul, América Central e Caribe.
- **Total stations:** 184 vértices geodésicos ocupados.
- **Stations in Brazil:** 21 estações continentais e insulares.

- **Three primary goals in 2000:**
  1. **Reoccupy 1995 stations:** measure crustal velocity and deformation vectors.
  2. **Tie tide gauges:** connect vertical datum origins to the geocentric ellipsoid.
  3. **Densify the network:** supply control marks within 500 km of any developed region.

The "Graduation Photo": Why Measure Simultaneously?

- **The school analogy:** if every graduate takes a photo alone on different days, how can you verify their exact relative height?
- In a class photo, **everyone steps on the risers together**: the same lens, the same lighting, the exact same moment!
- In satellite geodesy, GPS satellites orbit at $\approx 20200\,\text{km}$ altitude traveling at $3.9\,\textbf{km/s}$.
- **They act as a fast-moving space ruler**: they bridge two ground stations only when both track them at the exact same microsecond!

Principle of common tracking:

The atmosphere and the satellite's atomic clock drift continuously.

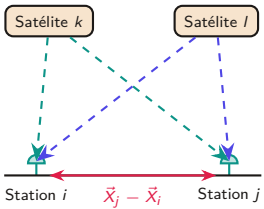
If Manaus and Curitiba record carrier phase from the same satellite at the exact same instant, the satellite clock error **is identical** for both and cancels out in subtraction!

The Double-Difference Phase Miracle

The carrier phase equation (Φ) between receiver i and satellite k:

$$\Phi_i^k = \rho_i^k + c(\delta t_i - \delta t^k) - I_i^k + T_i^k + \lambda N_i^k + \varepsilon$$

1. **First difference (between receivers):** $\Delta\Phi_{ij}^k = \Phi_j^k - \Phi_i^k$
   $\Longrightarrow$ **cancels satellite clock error** δt^k!
2. **Second difference (between satellites):** $\nabla\Delta\Phi_{ij}^{kl} = \Delta\Phi_{ij}^l - \Delta\Phi_{ij}^k$
   $\Longrightarrow$ **cancels receiver clock error** $\delta t_i, \delta t_j$!



Only pure geometry of the baseline vector and integer ambiguity N remain.

What Cancels and What Does NOT?

What cancels / is modeled

- **Clocks:** erro de relógio do satélite e do receptor (eliminados pelas diferenças).
- **Orbits:** produtos orbitais precisos finais do IGS (1 a 2 cm no espaço).
- **Ionosphere:** combinação livre da ionosfera L3 / L4 (combinação linear das frequências L1 e L2).

What does not cancel

- **Local troposphere:** vapor d'água na baixa atmosfera (exige estimar parâmetros de zênite a cada hora).
- **Multipath:** reflexão do sinal em construções, solo ou veículos vizinhos.
- **Antenna phase center:** variações mecânicas dependentes do ângulo zenital.

Notice: Over 2000 km baselines, the atmosphere over Manaus is completely different from Curitiba. Tropospheric delay must be solved per station!

Three Analysis Centers: Unbiased Science

- As in 1995, raw tracking data from May 2000 were processed by **three independent analysis centers**:
  1. **DGFI** (*Deutsches Geodätisches Forschungsinstitut*, Munique) — Bernese GPS Software 4.2;
  2. **IBGE** (Rio de Janeiro, Brasil) — Bernese GPS Software 4.0;
  3. **BEK** (*Bayerische Kommission für die Internationale Erdmessung*) — GIPSY/OASIS II.
- Notice: **two entirely different mathematical methodologies** (Bernese via double differences; GIPSY via PPP with Kalman filters).

Why multiple centers?

If a single center computes, a subtle code bug or solid Earth tide model error could taint a continent's mapping.

Blind comparison across independent solutions guarantees the output is **pure, reproducible physics**, not an algorithmic artifact.

The SINEX Format: Combining Equations Without Distortion

- Each center published not final rigid coordinates, but neutral **SINEX** files (*Solution INdependent EXchange*).
- The SINEX file contains:
  - A matriz de equações normais reduzida ($N = A^T PA$);
  - O vetor de termos independentes ($u = A^T Pl$);
  - A matriz de variância-covariância completa entre todas as estações.
- This enabled combining solutions as a unified mega-adjustment without rounding or covariance loss.

Least Squares Combination:

Given normal systems N_1, N_2, N_3 from the three centers with variance factors $\sigma_1^2, \sigma_2^2, \sigma_3^2$:

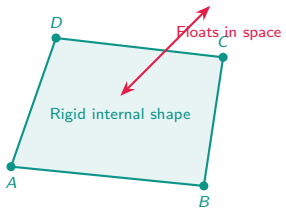
$$N_{\text{comb}} = \sum_{k=1}^{3} \frac{1}{\sigma_k^2} N_k$$

$$u_{\text{comb}} = \sum_{k=1}^{3} \frac{1}{\sigma_k^2} u_k$$

The combination is strictly optimal in the least-squares sense!

The Free Network and Rank Defect

- The combined GPS network determines relative baseline lengths and angles among 184 stations with **2 to 4 mm** precision.
- However, this perfect geometry is **unattached in space**:
  - Ela não sabe onde fica a origem $(0, 0, 0)$ do geocentro (defeito de translação de rank $= 3$);
  - A matriz de equações normais N é singular ($\det N = 0$) e não tem inversa direta!
- To convert this floating geometry into an official frame, we must **anchor the network to ITRS**.



Without anchors, the entire network can shift without altering any internal observation.

Adjustment with Constraints: The Mathematical Model

How to anchor the network to the global frame (ITRF2000):

We append observation equations for known coordinates of reference stations (IGS stations):

$$C\hat{x} = w + v_C, \quad \text{with weight matrix } P_C = \Sigma_C^{-1}$$

The normal equation system with weighted constraints becomes:

$$(\mathbf{N} + \mathbf{C^T P_C C})\hat{\mathbf{x}} = \mathbf{u} + \mathbf{C^T P_C w}$$

Minimal Constraint

Fixes only 3 translation parameters (Helmert condition). The internal network experiences zero artificial stress.

Weighted Constraint to ITRF

IGS coordinates enter with official uncertainties (Σ_C). The network ties to the geocenter with optimal fidelity.

Why is the Epoch 2000.4?

- The campaign took place between **May 10 and 19, 2000**.
- In the geodetic calendar, the campaign mean day is day of year 145 (May 24):

$$t = 2000 + \frac{145}{365{,}25} = \mathbf{2000.4}$$

- The global reference frame adopted was **ITRF2000**, propagated to epoch **2000.4**.
- The resulting coordinates represent the exact positions of stations **on that day of May 2000**!

The Frozen Frame Concept:

SIRGAS2000 was born as a *static frame*.

Unlike continuous ITRF (whose coordinates drift daily due to plate tectonics), SIRGAS2000 was **frozen at epoch 2000.4**.

This was essential so cadastral boundaries and national map sheets would not have to be redrawn every week!

The 21 Brazilian Stations in the 2000 Campaign

The 21 fundamental stations anchoring the new SGB across Brazil:

- **Norte:** Belém (PA), Macapá (AP), Manaus (AM), Oiapoque (AP), Porto Velho (RO), Rio Branco (AC).
- **Nordeste:** Fortaleza (CE), Imperatriz (MA), Natal (RN), Petrolina (PE), Salvador (BA), São Luís (MA).
- **Centro-Oeste:** Brasília (DF), Campo Grande (MS), Cuiabá (MT).
- **Sudeste:** Cananéia (SP), Presidente Prudente (SP), Rio de Janeiro (RJ), Viçosa (MG).
- **Sul:** Curitiba (PR), Santa Maria (RS).

In the 2006 network adjustment, **20 continental sites plus SMAR (Santa Maria)** served as the 21 constraint equations locking the entire country! [B]

The Cananéia curiosity:

Station CANA was one of 21 occupied in May 2000.

However, the mark was subsequently destroyed by local construction.

Therefore, in the 2006 calculation, IBGE replaced Cananéia with **SMAR** (Santa Maria), retaining exactly **21 constraint stations** across the nation! [B]

Tide Gauge Integration: Fulfilling the WG III Promise

- Recall: in 1997 at the IAG Assembly in Rio, SIRGAS formed **WG III (Vertical Datum)**.
- In May 2000, for the first time in Latin America, GPS antennas occupied marks adjacent to primary continental tide gauges:
  - Marégrafo de **Imbituba (SC)** — a origem altimétrica do SGB;
  - Marégrafo de **Santana (AP)** — a origem do datum vertical amapaense;
  - Marégrafos na Argentina, Chile, Colômbia, Venezuela e Uruguai.

Revolutionary significance:

Leveling yields height above sea level (H). GPS yields ellipsoidal height (h).

Surveying the tide gauge with GPS revealed with millimeter accuracy the **geoidal undulation** $N = h - H$ at the leveling datum origin!

This bridge enabled modern geoid modeling and vertical unification (the topic of our Part 2!).

Stop and Think: Why Did Brazil Wait After 2000?

Common doubt: "If the continental campaign finished in May 2000, why did Brazil wait until 2005 to decree SIRGAS2000 and 2006 to publish the national network adjustment?"

Gap between backbone and network

- Having 21 millimeter stations is merely the "backbone".
- Brazil had over 4700 **classical marks** on hilltops tied to SAD69.

The Herculean task (2000–2005)

- PMRG was created (2000) and 12 **GPS campaigns** ran in 2004–2005 tying classical marks.
- All 6265 marks had to be re-adjusted simultaneously without border seams! [B]

Your Turn: The Logic of Constraint Weights

Problem

In the 2006 network adjustment, IBGE constrained the 21 SIRGAS stations using a standard deviation divided by 10: $\sigma_{\text{inj}} = \sigma/10$.

(a) Since weight is proportional to $1/\sigma^2$, by what factor was the station weight multiplied?

(b) What was the practical consequence on the classical network stations?

Answers

(a) $P = \dfrac{1}{(\sigma/10)^2} = \mathbf{100} \times P_0$.

Weight increased by 100 **times**! [B]

(b) **Near-absolute stiffness:** the 21 SIRGAS stations act as unyielding pillars. The classical network is forced to fit them, preventing old distortions from pulling the geocentric frame!

Takeaways (Class 3)

1. The **May 2000** campaign simultaneously occupied 184 **stations** across two continents for 10 days of 24 hours, including 21 in Brazil.
2. Simultaneous tracking and **carrier double differences** eliminated satellite and receiver clock errors, yielding millimeter precision.
3. Three independent centers (DGFI, IBGE, BEK) processed data; normal equations in **SINEX** format were combined and adjusted with **weighted constraints to ITRF2000**.
4. The frame was frozen at epoch **2000.4**, also tying for the first time the physical leveling tide gauges (Imbituba, Santana).

Next class: legal adoption in Brazil (Decree 5,334/2005 and R.PR 1/2005), the 2006 network adjustment with 6,265 stations, conversion parameters, and frame lifetime.

Class 4: Adoption, Densification and Lifetime

Class 4 Hook: The Frozen Frame and the Moving Ground

The question for this class: once the 21 fundamental stations were computed in May 2000, how does that reach municipal maps, land titles, and field receivers? And if the frame was frozen at epoch **2000.4**, why does a modern receiver coordinate drift from the official monument datasheet?

- In Class 3 we saw the May 2000 campaign and the computation of the continental backbone.
- In this concluding class of Part 1, we will see how IBGE densified this accuracy across Brazil, the legal adoption framework, and living with tectonic plate motion.

The Densification Challenge: A Continental Nation

- Brazil covers over 8.5 **million** km^2. Twenty-one stations separated by thousands of kilometers cannot directly support daily land surveying.
- In 2000, IBGE formalized the **PMRG** (*Geodetic Network Modernization Program*), with two action fronts:
  1. **Active Network (RBMC):** continuous GNSS stations connected to the internet streaming data 24/7.
  2. **Passive Network Re-adjustment:** tie the thousands of classical concrete monuments to SIRGAS2000.

Birth of PMRG

Established at the 1st Seminar on the Brazilian Geodetic System (Curitiba, 2000), fulfilling the recommendation of the 1999 Brazilian Cartography Congress. [B]

The Active Network: RBMC

- The **RBMC** (*Brazilian Continuous Monitoring Network of GNSS Systems*) is the active backbone of Brazil.
- **Pioneering history:** inaugurated in **December 1996** with two pioneering stations:
  - Curitiba (UFPR, Paraná);
  - Presidente Prudente (UNESP, São Paulo).
- Today it comprises over one hundred stations providing hourly and daily RINEX files free of charge via IBGE.

The active network revolution:

Instead of hunting for a concrete monument hidden in brush, surveyors track with their own receiver and compute a baseline relative to the nearest RBMC station, whose official coordinate is known to the millimeter!

The IBGE-PPP Service: Democratizing High Precision

- The **IBGE-PPP** is a free public online service for processing GNSS data using Precise Point Positioning.
- Developed by IBGE in partnership with Natural Resources Canada (NRCan).
- **How it works:**
  1. O usuário envia um arquivo RINEX de rastreio estático pelo navegador;
  2. O servidor aplica órbitas e relógios precisos do IGS;
  3. O relatório PDF sai em minutos com coordenadas oficiais no **SIRGAS2000 na época 2000,4**.

Impact across Brazil:

Any citizen, municipality, or surveyor with a geodetic receiver gains direct access to IBGE's geodetic accuracy, without needing expensive commercial software packages.

The 2006 Planimetric Network Adjustment

- In June **2006**, IBGE concluded the largest least-squares adjustment in South American history. [B]
- **Adjustment scale:**
  - 6265 estações ajustadas simultaneamente;
  - 4774 marcos da rede clássica de triangulação/poligonação;
  - 1600 estações ocupadas por GPS moderno.
- **Total observations (22,232 equations):**
  - 16913 direções horizontais;
  - 1534 distâncias medidas;
  - 389 azimutes de Laplace;
  - 179 posições Doppler;
  - 3196 diferenças de coordenadas GPS;
  - 21 equações de injunção (estações SIRGAS com peso $\sigma/10$).

Final adjustment statistics:

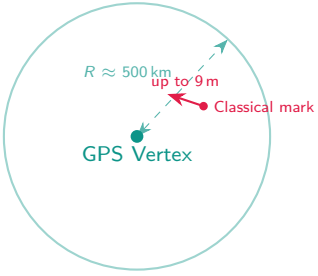
– Degrees of freedom: $\mathrm{DF} = \mathbf{9702}$;

– A posteriori variance factor: $\hat{\sigma}_0^2 = \mathbf{1.509}$;

– Partitioned into **8 Helmert blocks** solved without residual boundary seams across regions. [B]

GPS Connections: Clashing Old and New

- Between 2004 and 2005, IBGE carried out 12 **field GPS campaigns** reoccupying classical triangulation marks. [B]
- The result was startling: when an old mark received precise GPS coordinates, the adjustment shifted neighbors by up to 9 **m**, with an influence radius of 500 **km**! [B]
- This proved the classical SAD69 network harbored local elastic distortions from accumulated optical angle errors.



GPS acted not merely as a ruler, but as a rigorous auditor of the classical network.

Brazilian Legal Framework: Decree 5,334 and R.PR 1/2005

Federal Decree no. 5,334

January 6, 2005 (Presidency of the Republic)

Amended Article 21 of Decree 89,817/1984 (National Cartographic Standards):

"The planimetric and altimetric reference systems of national cartography are those of the **Brazilian Geodetic System**, *as defined by IBGE*". [E]

Resolution PR no. 01/2005

February 25, 2005 (IBGE Presidency)

– Decreed **SIRGAS2000** as Brazil's official reference frame (GRS80, geocentric).

– Repealed R.PR 22/1983 (which established SAD69).

– Established a **10-year transition period** allowing dual usage with older datums.

February 25, 2015: End of the Transition

- On **February 25, 2015**, the 10-year transition period established by Resolution PR 01/2005 officially expired.
- From that date forward:
  - **Córrego Alegre e SAD69** foram definitivamente extintos como referenciais legais;
  - Todo trabalho cartográfico, cadastral e de engenharia no Brasil deve ser compulsoriamente em **SIRGAS2000**;
  - O INCRA não aceita mais nenhuma planta de georreferenciamento de imóveis rurais (Lei 10.267) fora do SIRGAS2000.

Legal warning:

Delivering a boundary survey, subdivision plan, or engineering project in SAD69 today is a serious technical defect causing legal invalidity in land registries and environmental licensing!

The Conversion Formula: The 3 Official Parameters

To convert Cartesian coordinates from SAD69 to SIRGAS2000, IBGE determined in 2005 and reaffirmed in 2006 [B, tab. 9] three pure geocentric translation parameters:

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{\text{SIRGAS}} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{\text{SAD69}} + \begin{pmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{pmatrix}$$

- $\Delta X = -67.35\,\text{m}$;
- $\Delta Y = +3.88\,\text{m}$;
- $\Delta Z = -38.22\,\text{m}$.

Magnitude of the 3D shift vector:

$$|\Delta\vec{X}| = \sqrt{(-67{,}35)^2 + 3{,}88^2 + (-38{,}22)^2} \approx 77.54\,\text{m}$$

Why only 3 parameters?

IBGE tested 7-parameter Helmert models (3 rotations and scale) across 21, 63, and 113 common stations.

Rotations and scale were found statistically negligible nationally, while 7 parameters induced edge instabilities at border extremes. [B]

The “65-Meter” Myth: Real Shifts Vary by Region!

The 3D vector of 77.54 **m** has a fixed spatial orientation. Projected onto Earth’s curved surface, the horizontal shift varies from 57 to 76 **m**! (computed via `sgb_sirgas2000_numeros.py`):

City	*dN* (m)	*dE* (m)	*dh* (m)	Horiz.
Fortaleza	−41.65	−38.92	−29.50	57.00 **m**
Brasília	−49.54	−47.36	−12.87	68.54 **m**
Manaus	−40.11	−56.40	−11.88	69.21 **m**
Curitiba	−54.20	−48.50	−3.02	72.73 **m**
Chuí	−55.22	−51.80	+8.06	75.72 **m**
Cruzeiro do Sul	−40.88	−63.14	+4.51	75.22 **m**

Field warning:

Never apply a crude rule like “subtract 65 meters”!

In Fortaleza the horizontal shift is only 57.0 **m**, whereas in Chuí and Acre it exceeds 75 **m**.

Furthermore, ellipsoidal height (h) shifts from -29.5 **m** in the Northeast to +8.1 **m** in the South!

Grid Shift Method: ProGriD and NTv2

- The 3 parameters model only the shift between ellipsoid centers. But what about the **local distortions up to 13 m** in the old network?
- To transform legacy map sheets (1:25,000 and 1:50,000), IBGE developed **ProGriD**.
- Based on Canada's standard **NTv2** (*National Transformation version 2*):
  - Uma grade regular modela os deslocamentos locais $\delta\varphi$, $\delta\lambda$;
  - Interpolação bilinear suave na célula da grade;
  - Modela distorções locais que 3 parâmetros não conseguem absorver.

When to use which?

– **3 Parameters:** for modern GNSS surveys or general geographic conversions.

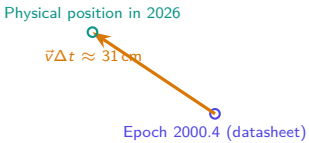
– **ProGriD (NTv2):** for converting digitized legacy maps surveyed from classical triangulation, where optical distortions are frozen in the original paper drafting.

The Crust Never Stops: Lifetime of SIRGAS2000

- Recall: SIRGAS2000 is **frozen at epoch 2000.4**.
- However, the South American Plate glides at about 1.2 **cm/yr** northwestward relative to Earth's deep mantle.
- Today (**2026**), 25.6 **years** have elapsed since the May 2000 campaign:

$$\Delta s \approx 1.2\,\mathrm{cm/ano} \times 25.6\,\mathrm{anos} \approx \mathbf{31\,cm}!$$

- If you turn on a geodetic GNSS receiver in the field today measuring in current ITRF, your physical mark sits about 31 **cm** away from the datasheet coordinate! [D]



The physical pillar traveled 31 cm riding the tectonic plate over the past 26 years.

Velocity Models: VEMOS and SIRGAS-CON

- To reconcile modern GNSS observations with the official frame frozen at 2000.4, SIRGAS maintains **VEMOS** (*Velocity Model for SIRGAS*).
- Model generations: VEMOS2003, VEMOS2009, VEMOS2015, and **VEMOS2022**, derived from over 400 continuous **SIRGAS-CON** stations.
- **Epoch reduction equation:**

$$\vec{X}(2000{,}4) = \vec{X}(t) - \vec{v} \cdot (t - 2000{,}4)$$

How IBGE-PPP handles this:

When you submit a RINEX file recorded yesterday to IBGE-PPP, the server solves your coordinate at the survey epoch and then applies crustal velocity models to output coordinates reduced to **2000.4**!

Modern Debate: Keep Frozen or Adopt Dynamic Datums?

Static Frame (Brazil / SIRGAS)

- **Advantage:** coordenadas de imóveis e mapas permanecem fixas por décadas. Ninguém precisa refazer escrituras a cada ano.
- **Challenge:** com o passar dos anos, a defasagem entre o GNSS de campo e o papel cresce (31 cm em 2026, 50 cm em 2040), exigindo modelos de velocidade.

Semi-Dynamic Frame (Australia)

- **Example:** a Austrália desloca-se a $\approx 7\,\text{cm/ano}$ para Nordeste. Em 2020 abandonou o GDA94 e lançou o **GDA2020** (1.8 m de salto!).
- **Challenge:** exige remapear todo o cadastro territorial nacional e mudar bases GIS a cada atualização de época.

Your Turn: Reducing a Coordinate to Official Epoch

Problem

A geodetic receiver surveyed in **2025.4** the X coordinate of a mark in ITRF as: $X(2025.4) = 2754120.350\,\mathrm{m}$.

(a) If plate velocity is $v_X = -0.010\,\mathrm{m/yr}$, what is the shift over 25 years?

(b) What is the official X coordinate for the SIRGAS2000 datasheet (epoch 2000.4)?

Step-by-step Solution

(a) $\Delta X = v_X \cdot (t - t_0)$

$= (-0.010\,\mathrm{m/ano}) \times (25\,\mathrm{anos}) =$ $\mathbf{-0.250}\,\mathrm{m}$.

(b) $X(2000{,}4) = X(t) - \Delta X$

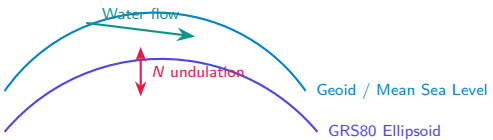
$= 2754120.350 - (-0.250)$

$= \mathbf{2754120.600}\,\mathrm{m}$.

Conclusion: A difference of $25\,\mathbf{cm}$ between survey epoch and official legal coordinate!

The Grand Bridge to Part 2: Water Doesn't Flow on the Ellipsoid!

- SIRGAS2000 flawlessly solved Brazil's 3D geometry: **latitude φ, longitude λ, and ellipsoidal height h**.
- However, in civil engineering, sanitation, canals, and dams, water does not follow the mathematical ellipsoid!
- Water flows along equipotential surfaces of **Earth's gravity field**:
  - Dois pontos com a mesma altura elipsoidal ($h_A = h_B$) podem ter altitudes físicas diferentes e a água escorrer de um para o outro!
- Physical height (H) comes from the **Vertical Leveling Network** tied to Imbituba Tide Gauge (SC).



The bridge between h (SIRGAS) and H (Imbituba) is the geoid model.

Part 1 Summary: The Four Classes at a Glance

Class	Core Idea of the Class
1	**The Heritage:** Classical datums (Córrego Alegre, SAD69) pinned ellipsoids to Minas Gerais (Chuá). Without geocenter, classical triangulation piled up 13 **m** distortion at border extremes.
2	**The Design:** In Asunción (1993), geodesists chose to anchor the frame to Earth's center of mass (ITRS, GRS80, GPS). SIRGAS95 demonstrated millimeter accuracy across 58 continental sites.
3	**The Realization:** In May 2000, 184 stations (21 in Brazil) tracked simultaneously. Processing in three independent centers and constrained adjustment yielded SIRGAS2000 frozen at epoch **2000.4**.
4	**Adoption and Lifetime:** Decree 5,334 and R.PR 1/2005 (transition ending 2015). The 2006 re-adjustment (6,265 stations). Plate drifts 1.2 **cm/yr** (31 cm in 2026), modeled by VEMOS.

Continues in Part 2: The Brazilian Geodetic System — The Leveling Network, Heights, and the Geoid!

Technical Appendix

Mathematical formulations, timeline and primary sources

A1 · ITRS and SIRGAS Definition

- **Origin:** Centro de massa da Terra total (incluindo oceanos e atmosfera).
- **Scale:** Determinada no sistema próprio de coordenadas relativísticas terrestres (TT / TCG).
- **Orientation:** Polo convencional terrestre (CIP) e meridiano de referência internacional (IRM), alinhados às convenções do IERS / BIH 1984.0.
- **Time evolution (NNR Condition):** A rotação líquida residual da crosta terrestre em relação ao manto profundo deve ser nula:

$$\int_{\text{Terra}} (\vec{r} \times \vec{v})\, dm = \vec{0}$$

- **Associated Ellipsoid:** GRS80 (*Geodetic Reference System 1980*):

$$a = \mathbf{6378137}\,\mathrm{m}, \quad 1/f = \mathbf{298.257222101}, \quad GM = 3986005 \times 10^8\,\mathrm{m}^3/\mathrm{s}^2$$

A2 · Least Squares Adjustment with Constraints

Given the linearized observation model $v = A\hat{x} - l$ with weights $P = \sigma_0^2 \Sigma^{-1}$, and constraint equations on known coordinates $v_C = C\hat{x} - w$ with weights $P_C = \sigma_0^2 \Sigma_C^{-1}$.

Combined Normal System

$$(A^T PA + C^T P_C C)\hat{x} = A^T Pl + C^T P_C w$$

Defining $N = A^T PA$ and $u = A^T Pl$:

$$\hat{x} = (N + C^T P_C C)^{-1}(u + C^T P_C w)$$

A Posteriori Variance

$$\hat{\sigma}_0^2 = \frac{v^T Pv + v_C^T P_C v_C}{(n + n_C) - u}$$

Where:

- n: número de observações de campo;
- n_C: número de equações de injunção;
- u: número de parâmetros incógnitos estimados.

A3 · Transformation Formulas SAD69 → SIRGAS2000

The official Brazilian transformation recommended by IBGE operates in 3D geocentric space:

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{\text{SIRGAS2000}} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{\text{SAD69}} + \begin{pmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{pmatrix}, \quad \begin{cases} \Delta X = -67.35\,\text{m} \\ \Delta Y = +3.88\,\text{m} \\ \Delta Z = -38.22\,\text{m} \end{cases}$$

Relationship with ellipsoidal parameters:

- SAD69 (UGGI67): $a_1 = 6378160\,\text{m}$, $1/f_1 = 298.25$;
- SIRGAS2000 (GRS80): $a_2 = 6378137\,\text{m}$ ($\Delta a = a_2 - a_1 = -23\,\text{m}$), $1/f_2 = 298.257222101$ ($\Delta f = -0.000000081$).
- Para conversão analítica direta em latitude e longitude geodésicas sem passar por ECEF, usa-se a fórmula de **Molodensky Abreviada** (implementada no simulador `sistemas_coordenadas/`).

A4 · Epoch Propagation with Plate Velocities

The 3D position of a station at epoch t is modeled via linear kinematics:

$$\vec{X}(t) = \vec{X}(t_0) + \vec{v} \cdot (t - t_0) + \sum_k \Delta\vec{X}_k \, H(t - t_k) + \vec{\varepsilon}(t)$$

- $\vec{X}(t_0)$: posição de referência na época oficial ($t_0 = 2000{,}4$);
- $\vec{v} = (v_X, v_Y, v_Z)^T$: velocidade linear constante da placa (fornecida pelos modelos VEMOS);
- $\Delta\vec{X}_k$: saltos co-sísmicos ou mudanças de antena no tempo t_k (H: função degrau Heaviside);
- $\vec{\varepsilon}(t)$: variações sazonais periódicas (carga hidrológica, marés oceânicas).

For practical users

To reduce today's survey (t) to official epoch $t_0 = 2000.4$:

$$\vec{X}(2000.4) = \vec{X}(t) - \vec{v} \cdot (t - 2000.4)$$

A5 · Brazilian Geodesy Timeline (1924–2026)

Year	Milestone	Technical Significance
1924	Hayford 1924	Adotado no datum Córrego Alegre ($a = 6378388\,\text{m}$, $1/f = 297$, $\xi = \eta = 0$).
1969	SAD69	UGGI67 ($a = 6378160\,\text{m}$, $1/f = 298.25$), datum no Vértice Chuá (MG).
1991	Chegada do GPS	IBGE adquire os primeiros 4 receptores geodésicos GPS de dupla frequência.
1993	Assunção	Conferência cria o SIRGAS (IAG, IPGH, DMA) com os Grupos de Trabalho I e II.
1995	SIRGAS95	Campanha com 58 estações continentais (11 no Brasil). ITRF94 época 1995,4.
1996	Ajuste SAD69 / RBMC	Ajuste global clássico (4760 estações); inauguração da RBMC (Curitiba e Pres. Prudente).
1997	GT III (Rio de Janeiro)	IAG cria o Grupo de Trabalho III (Datum Vertical e Altimetria).
2000	SIRGAS2000 / PMRG	Campanha de maio com 184 estações; época **2000,4**; criação formal do PMRG.
2005	Decreto 5.334 / R.PR 1	Oficialização legal do SIRGAS2000 no Brasil com período de transição de 10 anos.
2006	Ajuste da Rede Planimétrica	6.265 estações ajustadas simultaneamente com 21 injunções SIRGAS ($\sigma/10$).
2015	Fim da Transição	25/02/2015: encerramento definitivo do SAD69. Uso exclusivo de SIRGAS2000.
2026	Hoje	Crosta acumulou 31 cm de deriva; uso rotineiro de IBGE-PPP e modelos VEMOS2022.

A6 · Bilingual Conceptual Glossary

- **Datum Topocêntrico / Topocentric Datum:** referencial cujo elipsoide tangencia o geoide num ponto da superfície (Córrego Alegre, Chuá); centro não coincide com o geocentro.
- **Datum Geocêntrico / Geocentric Datum:** referencial com origem no centro de massa da Terra (ITRS, SIRGAS2000, WGS 84).
- **Sistema de Referência / Reference System:** definição teórica e física do espaço (regras, constantes a, f, GM, condição NNR).
- **Estrutura de Referência (Frame) / Reference Frame:** materialização física e numérica do sistema (lista de coordenadas e velocidades dos marcos em determinada época).
- **Rede Passiva / Passive Network:** marcos físicos cravados no solo (pilares, chapas de bronze) sem rastreamento contínuo.
- **Rede Ativa / Active Network:** estações com receptores GNSS permanentes transmitindo dados ininterruptos (RBMC).
- **Injunção / Constraint:** equação que fixa ou restringe com determinado peso as coordenadas de estações conhecidas no ajustamento.
- **Época de Referência / Reference Epoch:** data específica para a qual as coordenadas de um frame estático são calculadas e congeladas (ex.: 2000,4).

A7 · Key Acronyms and Institutions

Acronym	Name and Meaning
SGB	Brazilian Geodetic System (maintained and regulated by IBGE).
SIRGAS	Geodetic Reference System for the Americas.
ITRS / ITRF	*International Terrestrial Reference System / Frame* (IERS, padrão científico mundial).
IERS	*International Earth Rotation and Reference Systems Service* (Frankfurt / Paris).
IAG	*International Association of Geodesy* (International Association of Geodesy).
IPGH	Pan-American Institute of Geography and History (OAS specialized agency).
IGS	*International GNSS Service* (global satellite orbit and clock service).
DGFI-TUM	*Deutsches Geodätisches Forschungsinstitut* (Technical University of Munich, Germany).
DMA / NGA	*Defense Mapping Agency* (USA, later NIMA and now NGA).
IBGE	Brazilian Institute of Geography and Statistics.
RBMC	Brazilian Continuous Monitoring Network of GNSS Systems.
SINEX	*Solution INdependent EXchange* (neutral international format for exchanging geodetic normal equations).

A8 · Primary Sources and Fact Verification

- **[A]** IBGE (2000), *SIRGAS Final Report, Working Groups I and II*. Rio de Janeiro: IBGE / DGFI / NIMA. (Campanhas de 1995 e 2000, 58 e 184 estações, ITRF94 e ITRF2000).
- **[B]** IBGE (2006), *Adjustment of the Brazilian Planimetric Network in SIRGAS2000*. Rio de Janeiro: Diretoria de Geociências, Coordenação de Geodésia. (6.265 estações, 22.232 equações, parâmetros de translação da tab. 9, peso $\sigma/10$).
- **[C]** IBGE (1996/2002), *Adjustment of the SGB Planimetric Network*. Rio de Janeiro: IBGE. (Córrego Alegre, SAD69, histórico de triangulação, distorção de até 13 m no BDG).
- **[D]** SIRGAS (2026), `sirgas.ipgh.org`: Seções *About us*, *SIRGAS2000*, *Working Groups I, II, III*, *SIRGAS-CON*, *VEMOS Models*.
- **[E]** Presidency of the Republic of Brazil (2005), *Decree no. 5,334 of January 6, 2005*.
- **[F]** IBGE (2005), *Resolution PR no. 1/2005 of February 25, 2005*.
- Derived numbers and parameters recomputed in this repository: `sgb_sirgas2000_numeros.py`.

All facts, numbers, and legislation in this lecture were verified in full on 13/10/2026.